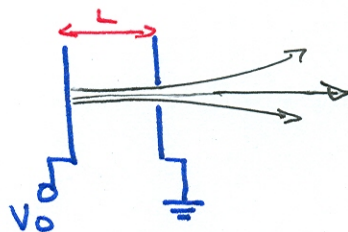
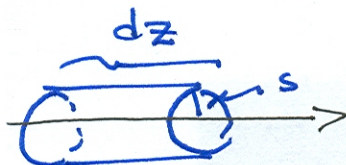


2.10 ELECTRON GUN



a) Consider a GAUSSIAN CYLINDER ALONG THE AXIS



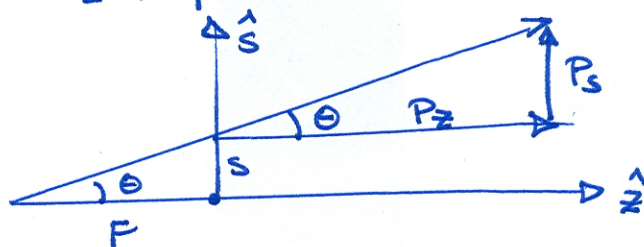
$$2\pi s dz E_s = \pi s^2 dE_z$$

$$\boxed{E_s(s) = \frac{1}{2} s \frac{dE_z}{dz}}$$

b) $dp_s = F_s dt = q E_s dt = \frac{1}{2} q s \frac{dE_z}{dz} dt = \frac{1}{2} q s dE_z \frac{1}{\beta c}$

$$P_{s \text{ FINAL}} = \int_0^{P_{s \text{ FINAL}}} dp_s = \frac{1}{2} q s \frac{1}{\beta c} \int_{E_{zi}}^{E_{zf}} dE_z$$

$$P_{s \text{ FINAL}} = \frac{1}{2} q s \frac{1}{\beta c} E_z$$



$$\frac{s}{F} = \frac{P_s}{P_z} \Rightarrow F = \frac{s P_z}{P_s}$$

$$E_z = \frac{V_0}{L} \quad P_z = \gamma \beta m_0 c$$

$$E_{\text{kin}} = q V_0 = (\gamma - 1) m_0 c^2 \Rightarrow V_0 = \frac{(\gamma - 1) m_0 c^2}{q}$$

$$F = \frac{s \gamma \beta m_0 c}{\frac{1}{2} q s \frac{1}{\beta c} E_z} = \frac{2 \gamma \beta^2 m_0 c^2}{q V_0 / L} = \frac{2 \gamma \beta^2 m_0 c^2 L}{q (\gamma - 1) m_0 c^2 / q}$$

$$\boxed{F = \frac{2 \gamma \beta^2}{(\gamma - 1)} L}$$

for small β

$$F = \frac{2 (1 + \frac{1}{2} \beta^2) \beta^2}{((1 + \frac{1}{2} \beta^2) - 1)} L = 4 (1 + \frac{1}{2} \beta^2) L$$

$$F = 4L$$

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} \approx 1 + \frac{1}{2} \beta^2 + \dots$$

FOR SMALL β